Topic 11: Database Design for Automation (SQLite, PostgreSQL & MySQL)

n8n Hands-On Intermediate Assessment - Level 2

1. Document Control

Field	Value
Document Title	Topic 11: Database Design for Automation Requirement Document
Version	1.0 (Approved)
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Prepared By	Lead Automation Architect (Antigravity AI / Tayana Academy)
Target Audience	Enterprise Automation Engineers, Database Administrators, System Integrators

2. Executive Summary & Objective

Build an enterprise relational transaction management workflow in n8n. The pipeline initializes relational database schemas with primary/foreign key constraints, safely inserts parent-child order records using parameterized queries, validates customer credit limits, executes automatic rollbacks upon constraint breaches, and retrieves consolidated order records via indexed SQL JOIN queries.

3. Architecture Diagram

```
[ Manual_Trigger ]
        |
        v
[ Code_InitRelationalDB ] (Initializes tables & seed data)
        |
        v
[ Code_StageIncomingOrder ] (Emits multi-item order)
        |
        v
[ Code_InsertOrderHeader ] (Inserts parent header: PENDING)
        |
        v
[ Code_InsertOrderItems ] (Inserts child items with Foreign Key)
        |
```

```
        v
[ Code_UpdateCustomerBalance ] (ACID constraint check: balance <= limit)
        |
        v
[ IF_TransactionSucceeded ] (isCommitted == true)
      /          \
(False)          (True)
    /              \
   v                v
[ Log_Rollback   [ Code_QueryFullOrderJoin ] (Indexed SQL JOIN)
  Incident ]         |
                     v
                [ Code_FinalTransactionAudit ] (Master Audit Log)
```

4. Node Configuration Blueprint

Node Name	Type	Operation / Setting	Output Outcome
Manual_Trigger	Manual Trigger	Trigger execution on demand	Execution pulse
Code_InitRelationalDB	Code	Initializes schema tables & customers	Database initialized schema
Code_StageIncomingOrder	Code	Emits test multi-item order	Order with lineItems array
Code_InsertOrderHeader	Code	Inserts parent order header (PENDING)	Header inserted with totalAmount
Code_InsertOrderItems	Code	Batch inserts child items with FK orderId	Child item records
Code_UpdateCustomerBalance	Code	ACID commit / rollback logic	isCommitted: true / false
IF_TransactionSucceeded	IF	isCommitted == true	Bifurcates commit vs rollback
Log_RollbackIncident	Code	Formats aborted transaction incident	Rollback audit log
Code_QueryFullOrderJoin	Code	Executes multi-table relational JOIN	Consolidated order + customer + items
Code_FinalTransactionAudit	Code	Master transaction receipt	Verified transaction log

5. Test Matrix & Verification Scenarios

Scenario	Customer & Balance	Order Total	Expected Outcome
Successful Commit	CUST-001 (Acme Corp) Balance: $1,200, Limit: $5,000	$1,275.00	Projected: $2,475 ≤ $5,000. Committed, order confirmed, JOIN executed, audit verified.
Credit Breach & Rollback	CUST-002 (Globex Inc) Balance: $450, Limit: $500	$1,275.00	Projected: $1,725 > $500. Rollback triggered, pending records purged, logged to Log_RollbackIncident.